Microeconomics I, Part B — TA Session I

Choice under Uncertainty

Selected exercises: 8, 9, 4, 10

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Exercise 8 (JD-QEM exam 2019–20). We consider an expected-utility decision-maker facing the following possible professional occupations: working in the financial industry (A), working in the movie industry (B), working in the car industry (C). We further assume that the decision-maker can apply to three different schools.

- After School 1, he is certain to find a job in the car industry.
- After School 2, he is certain to find a job in the financial industry.
- After School 3, he would find a job in the movie industry with probability 0.1 and in the financial industry with probability 0.9.

We assume that School 2 is the least preferred option of the decision-maker and that he is indifferent between School 1 and School 3.

(a) Give a representation of the utility function of this decision-maker.

(b) We assume that a new school (School 4) opens. After School 4, a student would find a job in each industry with probability $\frac{1}{3}$. How would the new school rank compared to Schools 1, 2, 3?

(c) There are too many applications to School 4. The ministry of education decides that rather than applying to School 4 directly, the decision-maker must apply to a lottery that leads to admission to School 4 with probability $\alpha \in [0, 1]$ and to admission to School 2 with probability $1 - \alpha$. For which values of α does the decision-maker prefer to apply to School 1 rather than to the lottery that may lead to admission to School 4?

Exercise 9 (JD-QEM exam 2020–21). We consider an expected-utility decision-maker with utility of the form $u(x) = x^a$ with $0 < a < 1$.

(a) Let X be a lottery whose outcome is uniformly distributed over $[0, 1]$. Determine $\mathbb{E}[u(X)]$.

(b) Let Y be a lottery whose outcome is 0 with probability $\frac{1}{3}$ and 1 with probability $\frac{2}{3}$. Determine $\mathbb{E}[u(Y)]$.

(c) Determine the value a^* (of a) for which the decision-maker is indifferent between X and Y.

(d) If $a > a^*$, which lottery, X or Y, is preferred by the decision-maker?

(e) Determine, as a function of a, the coefficient of absolute risk aversion of the decision-maker.

Exercise 4. Consider an investor who has to choose between two assets. Asset A has payoff $x_A = (12, 6, 9)$, all with equal probability. Asset B is risk free with payoff $x_B = (\bar{x}, \bar{x}, \bar{x})$.

(a) Assume $u(x) = \log(x)$. Find the certainty equivalent of x_A.

(b) Find the level of $\bar{x}$ such that the investor is indifferent between asset A and asset B.

(c) Assume now that $\bar{x} = \mathbb{E}[x_A]$. Find how many units h_A of asset A should be given to the agent to make him indifferent between h_A and one unit of the risk-free asset. Provide some verbal intuition.

(d) Take as reference point the expected payoff $\mathbb{E}[x_A] = 9$, around which the two extreme payoffs of asset A are symmetric, $12 = 9 + 3$ and $6 = 9 - 3$. Compute the probability premium $\pi(9, 3, u)$, that is, the π that solves

$$u(9) = \left(\frac{1}{2} + \pi\right) u(12) + \left(\frac{1}{2} - \pi\right) u(6).$$

Compare it with the gap between $\mathbb{E}[x_A]$ and the certainty equivalent found in (a): what do the two measures say about the investor's attitude towards risk?

(e) Compute the coefficient of absolute risk aversion.

Exercise 10 (Economics-QEM exam 2020–21). A decision maker has preferences over lotteries represented by an expected utility with Bernoulli utility $u(x)$. Consider the lottery that pays 0, 4 and 9 with probabilities $\frac{1}{6}$, $\frac{1}{2}$ and $\frac{1}{3}$:

$$L = \left\{(0, 4, 9); \left(\frac{1}{6}, \frac{1}{2}, \frac{1}{3}\right)\right\}.$$

(a) Provide the definition of the certainty equivalent of L given u.

(b) Compute the certainty equivalent of L given $u(x) = \sqrt{x}$.

(c) Compute the coefficient of absolute risk aversion given u and x.

(d) Assume that the agent starts with $w = 12$ and owns the lottery. Find the minimum price he is willing to accept to sell the lottery.

(e) Compare the minimum price found above and the certainty equivalent, and provide an intuition of their difference based on the coefficient of absolute risk aversion.